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Skew limiting bearing cage

Abstract

A rolling element bearing assembly includes a rolling element bearing assembly cage for spacing a plurality of rolling elements during operation of the rolling element bearing assembly about a central rotation axis. The cage includes a first axial end ring, a second axial end ring, and a plurality of bridges extending between the first axial end ring and the second axial end ring. The plurality of bridges defining roller pockets therebetween. At least one of the plurality of bridges includes a Rayleigh step array disposed on a surface thereof. The surface facing an interior of the roller pocket.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to U.S. Provisional Patent Application No. 63/175,886, filed on Apr. 16, 2021, the entire content of which is incorporated herein by reference.

FIELD OF THE INVENTION

(1) The present invention relates to bearing assemblies, and more particularly to a bearing assembly including a bearing cage configured limit rolling element skew.

BACKGROUND OF THE INVENTION

(2) The majority of bearing assemblies have no means to control the orientation of the rolling elements during operation. In some instances, high speed operation can cause bearings to skew, creating excess friction and heat, thus damaging the bearing assembly.

SUMMARY OF THE INVENTION

(3) The present invention provides, in one aspect, a rolling element bearing assembly cage for spacing a plurality of rolling elements during operation of a rolling element bearing assembly about a central rotation axis. The cage includes a first axial end ring, a second axial end ring, and a plurality of bridges extending between the first axial end ring and the second axial end ring. The plurality of bridges defining roller pockets therebetween. At least one of the plurality of bridges includes a Rayleigh step array disposed on a surface thereof. The surface facing an interior of the roller pocket.

(4) The present invention provides, in another aspect, a cage for spacing a plurality of rolling elements in a rolling element bearing assembly containing a working fluid. The cage includes a first axial end ring, a second axial end ring, a central rotation axis extending centrally through the first axial end ring and the second axial end ring, and a plurality of roller pockets spaced circumferentially about the cage. Each of the plurality of roller pockets is configured to receive a rolling element therein. At least one of the plurality of roller pockets includes a Rayleigh step array configured to selectively generate a hydrodynamic force within the working fluid in response to a skew angle between a rotational axis of the rolling element within the roller pocket and the central rotation axis.

(5) The present invention provides, in yet another aspect, a rolling element bearing assembly including an inner ring, an outer ring, a plurality of rolling elements disposed between the inner ring and the outer ring, and a cage disposed between the inner ring and the outer ring. The cage

includes a plurality of roller pockets in which the rolling elements are located. The plurality of roller pockets are configured to circumferentially space the rolling elements about the rolling element bearing assembly. The roller pockets include Rayleigh step arrays configured to align the rolling elements within the roller pockets.

(6) The present invention provides, in yet another aspect, a method of operating a rolling element bearing assembly. The method including providing an inner ring, an outer ring, a plurality of rolling elements disposed between the inner ring and the outer ring, and a cage configured to circumferentially space the plurality of rolling elements about the rolling element bearing assembly. The cage includes a plurality of roller pockets in which the rolling elements are located. Each roller pocket includes Rayleigh step arrays. The method further includes imparting relative rotation of the outer ring with respect to the inner ring with the rolling elements in rolling contact with the outer ring and the inner ring, and in response to skew of the rolling elements relative to a rotational axis of the rolling element bearing assembly, generating a hydrodynamic force on the rolling elements with the Rayleigh step arrays.

(7) Other features and aspects of the invention will become apparent by consideration of the following detailed description and accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is perspective view of a bearing assembly.

(2) FIG. 2 is a schematic view of a bearing assembly showing rolling element skew.

(3) FIG. 3 is a side view of one roller pocket of the bearing assembly of FIG. 1.

(4) FIG. 4 is a cross-sectional view taken along section line 4-4 in FIG. 3.

(5) FIG. 5A depicts a schematic side view of one rolling element experiencing skew in a first direction.

(6) FIG. 5B depicts a schematic side view of one rolling element experiencing skew in a second direction.

(7) FIG. 6 is a perspective view of a bearing cage, including a single rolling element, according to one embodiment of the present disclosure

(8) Before any embodiments of the invention are explained in detail, it is to be understood that the invention is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the following drawings. The invention is capable of other embodiments and of being practiced or of being carried out in various ways. Also, it is to be understood that the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting.

DETAILED DESCRIPTION

(9) FIG. 1 depicts a bearing assembly 4 including an inner ring 8, an outer ring (not shown), and a plurality of circumferentially distributed rolling elements 12 positioned between the inner and outer rings. The inner and outer rings share a common bearing axis A, which is the central rotational axis for the bearing assembly 4. Although a variety of applications are possible, the bearing assembly 4 provides low friction (i.e., rolling) support of a first component (e.g., a rotating shaft) within a second component (e.g., a stationary housing). The inner ring 8 includes a radially inner surface secured with the first component (not shown), and the outer ring includes a radially outer surface secured with the second component (not shown). Each of the rolling elements 12 defines a rolling surface 16 (FIG. 4) in contact with raceway surfaces of the inner and outer rings, respectively. The raceway surface 20 of the inner ring 8 is shown in FIG. 2. In the illustrated embodiment, the rolling elements 12 have tapered rolling surfaces 16, but other constructions (i.e., cylindrical) are optional. As shown in FIG. 1, a cage 24 extends circumferentially around the bearing axis A at a radial

position that is between the inner ring **8** and the outer ring such that the cage **24** occupies a common radial position with the plurality of rolling elements **12**. The cage **24** includes a pair of opposed axial rings **28, 32** and a plurality of bridges **36** extending generally axially between the two axial rings **28, 32**. In some embodiments, one or both axial rings **28, 32** may include a flange (e.g., a bent or rolled portion). A roller pocket **40** is defined between each adjacent pair of bridges **36**, and thus, the cage **24** maintains a predetermined spacing between adjacent pairs of the rolling elements **12** by locating a rolling element **12** within a roller pocket **40** (FIG. **6**).

(10) One or both of the inner and outer rings can include one or two ribs to retain the rolling elements **12** therein. For example, as shown in FIG. **2**, the inner ring **8** includes two ribs **44, 46** that extend radially outward beyond the raceway surface **20** to overlap with the axial end faces **48** of the plurality of rolling elements **12** on both axial ends of the bearing assembly **4**. The first rib **44** is referred to as the large rib as it is located at the larger diameter end of the bearing assembly **4**. The second rib **46** is referred to as the small rib as it is located at the smaller diameter end of the bearing assembly **4**. One of skill in the art will recognize that a variety of different rib configurations are optional. Whatever rib-roller interface exists, a potential for wear (e.g., metal-on-metal) exists as the bearing assembly **4** is in operation and the rolling elements **12** travel circumferentially with respect to the inner and outer rings. The interface between the large rib **44** and the rolling elements **12** experiences both rolling and sliding motion. Tractive forces generated at the rib-roller interface due to the combination of rolling and sliding motion can induce a skew angle θ on a rolling element **12** relative to the apex center A of the rolling element **12** and raceway **20** geometries (FIG. **2**). The skew angle θ of the rolling element **12** can cause a lack of lubrication at interface between the large rib **44** and the rolling element **12**, significantly increasing the frictional force associated with said interface. Conventional uses for tapered roller bearings typically operate in conditions where the effects of rolling element **12** skew are not a concern. In high speed operations, such as shaft positions in an electric vehicle powertrain, among others, the rotational speed required of a bearing assembly can surpass 15,000 RPM. At high rotational speeds, the tractive forces at the rib-roller interface can be great enough to cause the rolling element **12** to skew, starving the rolling element **12** of lubrication and damaging the bearing assembly **4**. However, the cage **24** according to the present disclosure is provided with a geometry to counteract rolling element **12** skew as described in further detail below.

(11) FIG. **3** depicts a rolling element **12** within one roller pocket **40** of a bearing assembly cage **24**. The roller pocket **40** is divided into four quadrants Q1, Q2, Q3, and Q4 by an axis B equidistant from and parallel to the cage bridges **36** and an axis C equidistant from and parallel to the opposed axial rings **28, 32** of the cage **24**. The two axes B, C intersect at a point on the rolling element **12** about which the rolling element **12** is rotated when experiencing skew. Within each of the four quadrants Q1, Q2, Q3, and Q4 is a Rayleigh step array **52** that is disposed on an interior face **56** of the bridge **36** that faces and selectively interacts with the rolling element **12**. The Rayleigh step arrays **52** are spaced away from the axis C. In other words, the cage bridge **36** features a Rayleigh step array **52** at each end of the interior face **56** and proximate to the respective opposed axial rings **28, 32**.

(12) A Rayleigh step array **52** is a grouping of one or more Rayleigh step geometries positioned together and oriented such that they are activated at the same time. The geometry of a Rayleigh step in this disclosure follows the general shape of a traditional Rayleigh step and is used to generate a hydrodynamic force F on a rolling element **12**. Dimensions of each feature of the Rayleigh step geometry can be tailored to the specific application. In the illustrated embodiment, each quadrant Q1, Q2, Q3, Q4 has one Rayleigh step array **52**, each containing multiple Rayleigh step geometries. FIG. **6** depicts a perspective view of a bearing cage **24** including Rayleigh step arrays **52** on the cage bridges **36**. In other embodiments, some quadrants can be provided without a Rayleigh step array or contain different numbers or arrangements of step geometries within the Rayleigh step array **52**. For example, a bearing assembly that will only rotate in one direction can

have Rayleigh step arrays **52** located in only one diagonal set of quadrants (e.g., the first and third quadrants **Q1**, **Q3** and no Rayleigh step arrays in the second and fourth quadrants **Q2**, **Q4**), because the bearing assembly will only require skew to be countered in one direction. Likewise, a roller pocket **40** may be provided with a Rayleigh step array in only one of the quadrants if deemed effective for a suitable application. Within each quadrant that contains a Rayleigh step array **52**, the step geometry is located such that the resultant hydrodynamic force generated when the arrays are activated acts with efficacy on the rolling element **12**. The specific location of each Rayleigh step in an array **52** can vary based on available manufacturing techniques and the desired direction of the resultant hydrodynamic force from the array **52**. The illustrated embodiment depicts the use of Rayleigh step arrays **52** on a single row tapered roller bearing assembly. However, the described Rayleigh step arrays **52** can be used on other bearing assemblies that utilize a cage and contains lubrication between the cage and rolling element surface. For example, cylindrical roller bearing assemblies, multi-row tapered roller bearing assemblies, and multi-row cylindrical roller bearing assemblies can all feature a cage with Rayleigh step arrays according to aspects of the disclosure.

(13) With respect to the rolling element **12** disposed within the roller pocket **40**, the Rayleigh step arrays **52** are located at positions on the bridge **36** that correspond to four quadrants **Q1'**, **Q2'**, **Q3'**, and **Q4'** on the surface of the rolling element **12**. The four rolling element quadrants **Q1'**, **Q2''**, **Q3''**, and **Q4'** are divided by a rotational centerline **D** of the rolling element **12** and an axis **E** parallel to the axial end faces **48** of the rolling element **12** and equidistant from said end faces **48**. Each quadrant **Q1'**, **Q2'**, **Q3''**, and **Q4'** is configured to interface with a Rayleigh step array **52** located in the corresponding quadrant **Q1-Q4** of the roller pocket **40** to selectively receive a hydrodynamic force **F** from the Rayleigh step arrays **52**. When a rolling element **12** is unskewed, the roller quadrants **Q1'-Q4'** are aligned with the roller pocket quadrants **Q1-Q4** and a clearance gap between the rolling element **12** and the surfaces **56** having the Rayleigh step arrays **52** is at or above a threshold value. When a rolling element **12** experiences a skew angle θ (FIGS. 5A and 5B) the quadrants of **Q1-Q4** are misaligned from the quadrants **Q1'-Q4'** and the clearance gap between the rolling element **12** and the cage surfaces **56** is below the threshold value within one diagonal quadrant pair. Below the threshold value, the Rayleigh step array **52** transitions from hydrodynamically dormant to hydrodynamically activated as described further below.

(14) FIG. 4 depicts a portion of one of the rolling elements **12** as viewed along the bearing axis **A**. As seen from this view, the Rayleigh step array **52** is located such that the hydrodynamic force **F** generated acts approximately tangent to the raceway **20** at the roller centerline **D**. The tangential component of the hydrodynamic force **F** generated by the Rayleigh step **52** is responsible for counteracting rolling element skew in the illustrated embodiment.

(15) The bearing assembly **4** of the present disclosure utilizes a running lubricant. Traditionally lubricants are used within bearing assemblies to limit the frictional forces associated with operation. Lubrication in the bearing assembly **4** of the present disclosure is also responsible for interacting with the Rayleigh step arrays **52** as a working fluid in which the hydrodynamic **F** is generated in response to rolling element skew. The Rayleigh step arrays **52** described herein are passive geometries, included on the bridges **36** of the cage **24**, that interact with the lubricant and a second surface (i.e., the rolling element surface **16**) to generate a hydrodynamic force **F**. Existence of the hydrodynamic force **F** is dependent upon the lubricant being disposed between the rolling elements **12** and the cage bridge surfaces **56** containing the Rayleigh step arrays **52**, and the relative velocity of the rolling elements **12** and Rayleigh step arrays **52**. Increased relative velocity between the rolling elements **12** and the Rayleigh step arrays **52** results in a greater hydrodynamic force **F** being generated.

(16) In operation, the Rayleigh step arrays **52** located within each roller pocket **40** act to prevent the rolling element **12** within the pocket **40** from experiencing prolonged skew. Moreover, the Rayleigh step arrays **52** are configured to be activated by roller skew and, once activated, to counteract the skew with a hydrodynamic force generated in the lubricant at the Rayleigh step arrays **52**. FIGS.

5A-5B schematically depict the two potential skew angles θ , θ' of a rolling element 12. A third rolling element position corresponds to a skew angle of zero and is not illustrated. The skew angle θ , θ' of the rolling element 12 is based on the direction of rotation of the bearing assembly 4. Only one direction of relative motion between a Rayleigh step array 52 and a rolling element 12 generates a hydrodynamic force F. To compensate for both potential directions of skew angle (i.e., angles θ , θ''), diagonally opposed Rayleigh step arrays 52 have the same geometric step directionality while adjacent Rayleigh step arrays 52 have opposite geometric step directionality. Thus, two adjacent Rayleigh step arrays 52 cannot be active at the same time—this would simply cause offsetting hydrodynamic forces F and the skew θ , θ' will not be compensated for. A rolling element 12 that has a first skew angle θ (FIG. 5A) activates the Rayleigh step arrays 52 in the second and fourth quadrants Q2, Q4 while leaving the Rayleigh step arrays 52 in the first and third quadrants Q1, Q3 deactivated. The hydrodynamic forces F from the second and fourth quadrant Rayleigh step arrays 52 combine to create a force couple acting on the rolling element 12 to counter the skew angle θ . The activated Rayleigh step arrays 52 may reduce the skew angle to an acceptable amount or eliminate the skew angle θ altogether. If the direction of rotation of the bearing assembly 4 is reversed, causing the rolling element 12 to experience a skew angle θ' (FIG. 5B), the second and fourth quadrant Q2, Q4 Rayleigh step arrays 52 remain deactivated while the first and third quadrant Q1, Q3 Rayleigh step arrays 52 are activated. If the rolling element 12 experiences little or no skew the Rayleigh step arrays 52 in all four quadrants Q1, Q2, Q3, and Q4 simply remain deactivated and the cage bridges 36 function in a conventional manner to maintain spacing of the rolling elements 12. The hydrodynamic force F generated by a Rayleigh step array 52 is a function of the relative velocity of the rolling element 12 and the Rayleigh step array 52. As the relative velocity increases, due to a greater rotational speed of the bearing assembly 4, the Rayleigh step array 52 generates a greater hydrodynamic force F. This effect allows the Rayleigh step array 52 to compensate for different magnitudes of rolling element skew based on the rotational speed of the bearing assembly 4. A bearing assembly with a high rotational speed will induce greater skew on the rolling elements 12. At the same time, the higher rotational speed will generate a greater hydrodynamic force from the active Rayleigh step arrays 52. Thus, the design of the cage 24 with at least one Rayleigh step array may render a bearing assembly that is otherwise unsuitable or unable to be rated for high speed operation, now suitable or rated for high speed operation. One such example is the use of a tapered rolling element bearing within an electric vehicle drivetrain.

(17) Although the invention has been described in detail with reference to certain preferred embodiments, variations and modifications exist within the scope and spirit of one or more independent aspects of the invention as described.

(18) Various features of the invention are set forth in the following claims.

Claims

1. A rolling element bearing assembly cage for spacing a plurality of rolling elements during operation of the rolling element bearing assembly about a central rotation axis, the cage comprising: a first axial end ring; a second axial end ring; and a plurality of bridges extending between the first axial end ring and the second axial end ring, the plurality of bridges defining a corresponding plurality of roller pockets therebetween, at least one of the plurality of bridges including a Rayleigh step array disposed on a surface thereof, the surface facing an interior of the roller pocket.
2. The cage of claim 1, wherein the roller pocket which the Rayleigh step array faces includes four quadrants, wherein the Rayleigh step array is disposed within a first of the quadrants, and wherein each of the other three quadrants includes a Rayleigh step array.
3. The cage of claim 1, wherein each of the plurality of roller pockets is tapered, and wherein the

plurality of tapered roller pockets are configured to receive tapered rolling elements therein.

4. The cage of claim 1, wherein each of the plurality of roller pockets includes a Rayleigh step array.

5. A rolling element bearing assembly comprising a cage according to claim 1, the rolling element bearing assembly further comprising: an inner ring; an outer ring; and a plurality of rolling elements situated between the inner and outer rings and spaced circumferentially from each other by the plurality of bridges of the cage, the plurality of rolling elements configured to facilitate relative rotation between the inner ring and the outer ring.

6. A cage for spacing a plurality of rolling elements in a rolling element bearing assembly containing a working fluid, the cage comprising: a first axial end ring; a second axial end ring; a central rotation axis extending centrally through the first axial end ring and the second axial end ring; and a plurality of roller pockets spaced circumferentially about the cage, each of the plurality of roller pockets configured to receive a rolling element therein, wherein at least one of the plurality of roller pockets includes a Rayleigh step array that generates a hydrodynamic force within the working fluid in response to a skew angle between a rotational axis of the rolling element within the roller pocket and the central rotation axis.

7. The cage of claim 6, wherein the roller pocket having the Rayleigh step array includes four quadrants, and wherein each of the four quadrants includes a Rayleigh step array.

8. The cage of claim 7, wherein the Rayleigh step arrays in adjacent quadrants have opposite geometric step directionality.

9. The cage of claim 8, wherein each Rayleigh step array includes a plurality of distinct Rayleigh step geometries.

10. The cage of claim 9, wherein each Rayleigh step geometry within a Rayleigh step array has the same geometric step directionality.

11. The cage of claim 6, wherein each of the plurality of roller pockets includes a Rayleigh step array.

12. A rolling element bearing assembly comprising a cage according to claim 6, the rolling element bearing assembly further comprising: an inner ring; an outer ring; and a plurality of rolling elements disposed within the plurality of roller pockets, the plurality of rolling elements configured to facilitate relative rotation between the inner ring and the outer ring.

13. The rolling element bearing assembly of claim 12, wherein the plurality of rolling elements are tapered rolling elements, and wherein the plurality of roller pockets are tapered roller pockets.

14. The rolling element bearing assembly of claim 12, wherein each of the plurality of roller pockets includes a Rayleigh step array.

15. The rolling element bearing assembly of claim 12, wherein a first rolling element of the plurality of rolling elements is disposed within the at least one of the plurality of roller pockets, the Rayleigh step array is a first Rayleigh step array of a plurality of Rayleigh step arrays, the first Rayleigh step array is configured to generate a first hydrodynamic force on the first rolling element when the outer ring is rotated relative to the inner ring in a first direction, and a second Rayleigh step array of the plurality of Rayleigh step arrays is configured to generate a second hydrodynamic force on the first rolling element when the outer ring is rotated relative to the inner ring in a second direction, opposite the first direction, the second hydrodynamic force acting in a different direction than the first hydrodynamic force.

16. The rolling element bearing assembly of claim 15, wherein the first hydrodynamic force acts on the first rolling element in a different location than the second hydrodynamic force.

17. A method of operating a rolling element bearing assembly, the method comprising: providing an inner ring, an outer ring, a plurality of rolling elements disposed between the inner ring and the outer ring, and a cage configured to circumferentially space the plurality of rolling elements about the rolling element bearing assembly, the cage including a plurality of roller pockets in which the plurality of rolling elements are located, at least one of the plurality of roller pockets including a

Rayleigh step array; imparting relative rotation between the outer ring and the inner ring to set the plurality of rolling elements in rolling contact with the outer ring and the inner ring, the rotation imparting skew to at least one of the plurality of rolling elements; and in response to the skew of the at least one of the plurality of rolling elements, generating a skew counteracting hydrodynamic force on at least one of the plurality of rolling elements with the Rayleigh step array.

18. The method of claim 17, wherein each of the plurality of roller pockets includes a Rayleigh step array, each of the Rayleigh step arrays generating a skew counteracting hydrodynamic force on the plurality of rolling elements in response to the skew of the plurality of rolling elements.

19. The method of claim 17, wherein the outer ring includes a rib configured to secure the plurality of rolling elements between the outer ring and the inner ring, the rib imparting the skew to the plurality of rolling elements during rotation.
